- VISCOUS FORCE $\sim \mu\, du/dy \cdot AREA \sim \mu V/L \cdot L^2 \sim \mu VL$
  PRESSURE FORCE $\sim pL^2 \sim \Delta p L^2$
  GRAVITY FORCE $\sim mg \sim \rho L^3 g$
  SURFACE TENSION FORCE $\sim \sigma L$
  INERTIAL FORCE $\sim m.a \sim \rho L^3 V^2/L \sim \rho V^2 L^2$

REYNOLDS NO. $\sim$ INERTIAL/VISCOUS $\sim \rho V^2L^2/\mu VL = \rho VL/\mu$
FROUDE NO.$^2 \sim$ INERTIAL/GRAVITY $\sim \rho V^2L^2/\rho L^3 g = V^2/gL$
WEBER NO. $\sim$ INERTIAL/SURFACE TENSION $\sim \rho V^2L^2/\sigma L = \rho V^2 L/\sigma$
→ MAY APPEAR IN A DIFFERENT FORM, BUT IT ALWAYS CONVEYS THE SAME MEANING !!

EXAMPLE: THE DRAG OF A SONAR TRANSDUCER IS TO BE PREDICTED, BASED ON WIND TUNNEL TEST DATA. THE PROTOTYPE, A 0.3m DIAMETER SPHERE, IS TO BE TOWED AT 2.57 m/s IN SEAWATER AT 4.5°C. THE MODEL IS 152mm IN DIAMETER. DETERMINE THE REQUIRED TEST SPEED IN AIR. IF THE DRAG OF THE MODEL UNDER THESE TEST CONDITIONS IS 2.7N, ESTIMATE THE DRAG OF THE PROTOTYPE.

FIRST, WE NEED TO DEVELOP A FORMULA FOR DRAG FORCE USING ESTIMATES FROM THE BUCKINGHAM-PI THEOREM. IT IS REASONABLE TO EXPECT THAT DRAG DEPENDS ON ρ, μ, V, D. THUS, WE HAVE $g(F_D, \rho, \mu, V, D) = 0$. TAKING ρ, μ, D AS OUR PRIMARY VARIABLES, WE GET.

$[\rho] = ML^{-3}$, $[\mu] = ML^{-1}T^{-1}$, $[D] = L$ $\Rightarrow \pi_1 = F_D \rho^a \mu^b D^c \Rightarrow \pi_1 = (MLT^{-2})(ML^{-3})^a (ML^{-1}T^{-1})^b L^c = [0]$

$\Rightarrow M^{1+a+b} L^{1-3a-b+c} T^{-2-b} = [0]$

$\Rightarrow 1+a+b=0,\ 1-3a-b+c=0,\ -2-b=0 \Rightarrow b=-2, a=1, c=0$

$\Rightarrow \pi_1 = F_D \cdot \rho \mu^{-2}$

SIMILARLY, $\pi_2 = V \rho^a \mu^b D^c \Rightarrow \pi_2 = (LT^{-1})(ML^{-3})^a (ML^{-1}T^{-1})^b L^c = [0]$

$\Rightarrow M^{a+b} L^{1-3a-b+c} T^{-1-b} = [0] \Rightarrow b=-1, a=1, c=1$

$\Rightarrow \pi_2 = V\rho D/\mu$

HENCE, $G\left(\frac{F_D \rho}{\mu^2}, \frac{\rho VD}{\mu}\right) = 0$, $G\left(\frac{F_D \rho}{\mu^2}, \frac{\rho VD}{\mu}\right) = 0$

FOR GEOMETRIC SIMILARITY, $Re|_{PROTOTYPE} = Re|_{MODEL}$

SEAWATER @ 4.5°C $\Rightarrow \rho = 1000\ kg/m^3$, $\mu = 1.57 \times 10^{-3}\ m^2/s \Rightarrow Re|_{PROTOTYPE} = \frac{1000 \times 2.57 \times 0.3}{1.57 \times 10^{-3}} = 4.9 \times 10^5$

AIR AT STP $\Rightarrow \rho = 1.227\ kg/m^3$, $\nu = 1.46 \times 10^{-5}\ m^2/s \Rightarrow 4.91 \times 10^5 = \frac{0.152 \times V}{1.46 \times 10^{-5}} \Rightarrow V = 47.16\ m/s$. (TEST SPEED IN AIR)

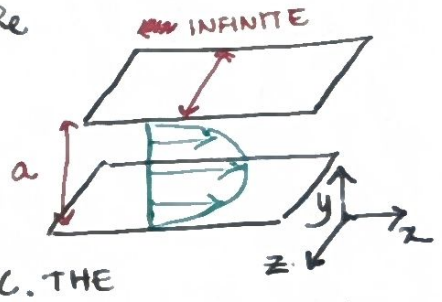
FOR THESE CONDITIONS, $\left.\frac{F_D \rho}{\mu^2}\right|_{MODEL} = \left.\frac{F_D \rho}{\mu^2}\right|_{PROTOTYPE} \Rightarrow F_D|_{PROTOTYPE} = 25.5\ N$

• TURBULENT FLOW IS INHERENTLY CHAOTIC, STEADY ASSUMPTION IS NOT APPLICABLE.

- FOR LAMINAR FLOWS, ENTRANCE LENGTH VARIES APPROXIMATELY AS $L/D \approx 0.06 \cdot Re$
- FULLY DEVELOPED LAMINAR FLOW BETWEEN INFINITE PARALLEL PLATES:

INCOMPRESSIBLE NAVIER STOKES: $\partial v/\partial y = 0$, $\mu \partial^2 u/\partial y^2 = \partial p/\partial x$, $-\rho g = \partial p/\partial y$

$u = \frac{a^2}{2\mu}\left(\frac{\partial p}{\partial x}\right)\left\{\left(\frac{y}{a}\right)^2 - \left(\frac{y}{a}\right)\right\}$, $\tau = \mu \frac{\partial u}{\partial y} = a \frac{\partial p}{\partial x}\left\{\frac{y}{a} - \frac{1}{2}\right\}$, $Q/\ell = -\frac{1}{12\mu}\left(\frac{\partial p}{\partial x}\right) a^3$, $\bar{V} = \frac{Q}{A}$

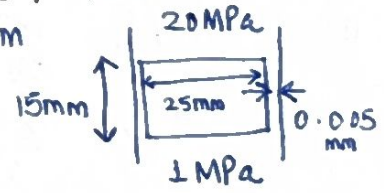


EXAMPLE: A HYDRAULIC SYSTEM OPERATES AT A GAGE PRESSURE OF 20 MPa AND 55°C. THE HYDRAULIC FLUID IS SAE 10W OIL. A CONTROL VALVE CONSISTS A PISTON 25 mm IN DIAMETER, FITTED TO A CYLINDER WITH A MEAN RADIAL CLEARENCE OF 0.005 mm. DETERMINE THE LEAKAGE FLOW RATE IF THE GAGE PRESSURE ON THE LOW PRESSURE SIDE OF THE PISTON IS 1.0 MPa. THE PISTON IS 15mm LONG.

BY FORMULA, $Q/\ell = \frac{a^3 \cdot \Delta p}{12 \mu L}$, WHERE $\Delta p = 19\ MPa$, $L = 15mm$, $a = 0.005mm$, $\ell = \pi \times 25mm$

μ (SAE OIL) $= 0.018\ kg/(m \cdot s)$

$\Rightarrow Q = 576\ mm^3/s$.

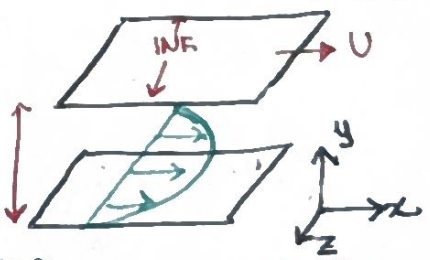


• CONSIDER CONFIGURATION WHERE TOP PLATE MOVES WITH VELOCITY U.

- COM: $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$, COLM-x: $\rho\left\{\frac{\partial u}{\partial t}^{0} + u\frac{\partial u}{\partial x}^{0} + v\frac{\partial u}{\partial y}\right\} = -\frac{\partial p}{\partial x} + \mu\left\{\frac{\partial^2 u}{\partial x^2}^{0} + \frac{\partial^2 u}{\partial y^2}\right\}$ (FROM COM)

(^{0} ALSO, $\partial v/\partial x = 0$ $\Rightarrow v = CONST. = 0$ CBC)

COLM-y: $\rho\left\{\frac{\partial v}{\partial t}^{0} + u\frac{\partial v}{\partial x}^{0} + v\frac{\partial v}{\partial y}^{0}\right\} = -\frac{\partial p}{\partial y} + \mu\left\{\frac{\partial^2 v}{\partial x^2}^{0} + \frac{\partial^2 v}{\partial y^2}^{0}\right\} - \rho g$

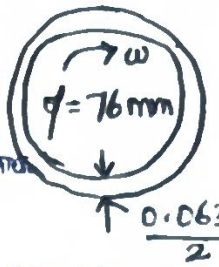


$\Rightarrow \mu \frac{\partial^2 u}{\partial y^2} = \frac{\partial p}{\partial x}$, $\frac{\partial p}{\partial y} = -\rho g \Rightarrow p = p_0 - \rho g (y - y_0) + F(x) \Rightarrow \frac{\partial p}{\partial x} = F'(x)$ → JUST FUNCTION OF x.

(JUST FUNCTION OF y; JUST FUNCTION OF x)

$\Rightarrow u(y) = \frac{1}{\mu} \cdot \frac{\partial p}{\partial x} \cdot \frac{y^2}{2} + C_1 y + C_2$, s.t. @ $y=0, u=0$; @ $y=a, u=U$ $\Rightarrow u(y) = \underbrace{\frac{Uy}{a}}_{\text{COUETTE}} + \underbrace{\frac{a^2}{2\mu}\left(\frac{dp}{dx}\right)\left\{\left(\frac{y}{a}\right)^2 - \left(\frac{y}{a}\right)\right\}}_{\text{POISEUILLE}}$

EXAMPLE: A CRANKSHAFT JOURNAL BEARING IN AN AUTOMOBILE ENGINE IS LUBRICATED BY SAE OIL AT 99°C. BEARING DIAMETER IS 76 MM, DIAMETRICAL CLEARENCE IS 0.0635mm, SHAFT ROTATES AT 3600 RPM; IT IS 31.8 MM LONG. BEARING ISUNDER NO LOAD, SO CLEARENCE IS SYMMETRIC. DETERMINE TORQUE REQD. TO TURN JOURNAL, AND POWER DISSIPATED.



GAP WIDTH IS MUCH SMALLER THAN DIAMETER, SO WE CAN APPROXIMATE AS COUETTE FLOW B/W PLATES.

ALSO, $\partial p/\partial x = 0 \Rightarrow \tau_{yx} = \mu \frac{U}{a}$, WHERE $U = \omega \cdot D/2 \Rightarrow \tau_{yx} = \mu \cdot \frac{\omega D}{2a}$

FOR GIVEN CONDITIONS, $\mu_{OIL} = 9.6\times10^{-3} \Rightarrow \tau_{yx} = 9.6\times10^{-3} \times 3600 \times \frac{2\pi}{60} \times \frac{76}{1000} \times \frac{1\times1000}{2\times0.0635} = 4331.6\ Pa$

$\Rightarrow$ SHEAR FORCE = $\tau_{yx} \times$ AREA = $\tau_{yx} \times \pi D \times L \Rightarrow$ TORQUE = $\tau_{yx} \times \pi D \times L \times D/2 = 1.25\ Nm$.

$\therefore$ POWER = TORQUE $\times \omega = 471\ W$.

- COM: $\frac{1}{r}\frac{\partial}{\partial r}(r u_r) + \frac{1}{r}\frac{\partial u_\theta}{\partial \theta} + \frac{\partial u_z}{\partial z} = 0$ (AXISYMMETRIC; FULLY DEVELOPED)

$\Rightarrow \frac{1}{r}\frac{\partial}{\partial r}(r u_r) = 0 \Rightarrow r u_r = F(z)$

@ $r=0, u_r=0 \Rightarrow F(z)=0 \Rightarrow r u_r = 0$ (ALWAYS TRUE) $\Rightarrow u_r = 0$

@ $r=R, u_r=0 \Rightarrow F(z)=0$

$u_r=0, u_z=0$ — SYMMETRY — $u_r=0, \partial u_z/\partial r = 0$

(NOT ASSUMPTIONS, FOLLOWS FROM B.C.S)

- COLM-r: $\rho\left\{\frac{\partial u_r}{\partial t} + u_r\frac{\partial u_r}{\partial r} + \frac{u_\theta}{r}\frac{\partial u_r}{\partial \theta} + u_z\frac{\partial u_r}{\partial z} - \frac{u_\theta^2}{r}\right\} = -\frac{\partial p}{\partial r} + \rho g_r + \mu\left\{\nabla^2 u_r - \frac{u_r}{r^2} - \frac{2}{r^2}\frac{\partial u_\theta}{\partial \theta}\right\}$

(0 AS AXISYMMETRY; AXISYMMETRY)

$\Rightarrow \frac{\partial p}{\partial r} = \rho g_r \Rightarrow p = p_0 + \rho g_r (r - r_0) + G(z)$ } CONCLUSIONS FROM HERE

$\Rightarrow \partial p/\partial z = G'(z) \rightarrow$ PURELY FUNCTION OF z.

- COLM-z: $\rho\left\{\frac{\partial u_z}{\partial t} + u_r\frac{\partial u_z}{\partial r} + \frac{u_\theta}{r}\frac{\partial u_z}{\partial \theta} + u_z\frac{\partial u_z}{\partial z}\right\} = -\frac{\partial p}{\partial z} + \mu\left\{\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_z}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u_z}{\partial \theta^2} + \frac{\partial^2 u_z}{\partial z^2}\right\}$

(STEADY; 0; AXISYMMETRY; DEVELOPED; AXISYMMETRY; DEVELOPED)

$\Rightarrow \frac{\partial p}{\partial z} = \boxed{\frac{\mu}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_z}{\partial r}\right)}$ → ONLY VARIES WITH r = CONSTANT

ONLY VARIES WITH z

$\Rightarrow \frac{r}{\mu}\frac{\partial p}{\partial z} = \frac{\partial}{\partial r}\left(r\frac{\partial u_z}{\partial r}\right) \Rightarrow r\frac{\partial u_z}{\partial r} = \frac{r^2}{2\mu}\cdot\left(\frac{\partial p}{\partial z}\right) + C_1$

$\Rightarrow \frac{\partial u_z}{\partial r} = \frac{r}{2\mu}\left(\frac{\partial p}{\partial z}\right) + C_1/r$

@ $r=0, \partial u_z/\partial r = 0 \Rightarrow C_1 = 0$

HENCE, $\frac{\partial u_z}{\partial r} = \frac{r}{2\mu}\cdot\left(\frac{\partial p}{\partial z}\right) \Rightarrow u_z = \frac{r^2}{4\mu}\cdot\left(\frac{\partial p}{\partial z}\right) + C_2$

@ $r=R, u_z=0 \Rightarrow -\frac{R^2}{4\mu}\cdot\left(\frac{\partial p}{\partial z}\right) = C_2 \Rightarrow u_z = -\frac{R^2}{4\mu}\cdot\left(\frac{\partial p}{\partial z}\right)\left\{1 - \left(\frac{r}{R}\right)^2\right\}$

EXAMPLE: WATER AT 20°C FLOWS IN A 4.3 cm DIAMETER PIPE THAT IS 70m LONG AND DISCHARGES TO THE ATMOSPHERE. PRESSURE AT ENTRANCE IS 1.82 KPa GAGE. DETERMINE MASS FLOW RATE & REYNOLDS NUMBER.

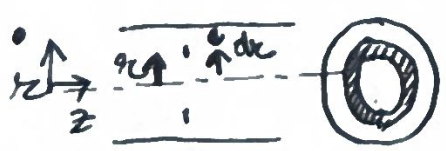
BY FORMULA, $Q = \frac{\pi \Delta p D^4}{128 \mu L}$ (ASSUMPTION IS THAT FLOW BECOMES FULLY DEVELOPED RIGHT AT ENTRY WHERE PRESSURE IS SPECIFIED)

$\Rightarrow Q = \frac{\pi \times 1.82\times10^3 \times (4.3\times10^{-2})^4}{128\times1.01\times10^{-3}\times70} = 2.16\times10^{-3}\ m^3/s \Rightarrow \bar{v} = \frac{Q}{A} = \frac{2.16\times10^{-3}\times4}{\pi\times(4.3\times10^{-2})^2} = 1.48\ m/s$

$\Rightarrow Re = \frac{\rho V D}{\mu} = \frac{998\times1.48\times4.3\times10^{-2}}{1.01\times10^{-3}} \approx 62883$ (LAMINAR FLOW ASSUMPTION IS INVALID).

$\Rightarrow$ ANALYTICAL FORMULA CANNOT BE USED!!

\#



FORCE BALANCE:

$\tau \cdot 2\pi r dz$; $p \cdot 2\pi r dr$; $\left(p + \frac{\partial p}{\partial z}\cdot dz\right) 2\pi r dr$; $\left\{\tau + \frac{\partial \tau}{\partial r}\cdot dr\right\} 2\pi (r+dr) dz$

$\Rightarrow -\frac{\partial p}{\partial z}\cdot 2\pi r dr dz + \tau \cdot 2\pi dr dz + \frac{\partial \tau}{\partial r}\cdot 2\pi r dr dz = 0$

$\Rightarrow r\frac{\partial p}{\partial z} = \frac{d}{dr}(r\tau) \Rightarrow \tau = \frac{r}{2}\frac{dp}{dz}$ } AFTER REMOVING CONSTANT OF INTEGRATION!!

- PRANDTL MIXING LENGTH THEORY: $u' = -l\,d\bar{u}/dy$, $l \equiv$ MIXING LENGTH.
- FLUCTUATION IN VERTICAL COMPONENT, $v' = l\,d\bar{u}/dy$ (HOMOGENEOUS TURBULENCE)

$\Rightarrow \frac{\tau_{TUR.}}{\rho} = l^2\left(\frac{d\bar{u}}{dy}\right)^2$

- MIXING LENGTH INCREASES WITH DISTANCE FROM WALL $\Rightarrow l = ky$
- WALL SHEAR STRESS SUB-REGION, LAMINAR DOMINANT $\Rightarrow \tau_w/\rho = \nu\, d\bar{u}/dy$: DEFINE $u_* = \sqrt{\tau_w/\rho}$, $u_+ = \bar{u}/u_*$, $y_+ = y u_*/\nu$

$\Rightarrow u_*^2 = \nu \frac{d(u+u_*)}{d(\nu y_+)} \cdot u_*$

$\Rightarrow du_+ = dy_+ \Rightarrow u_+ = y_+$ } WALL SHEAR + LAMINAR DOMINATES, $y_+ < 5$

- WALL SHEAR STRESS SUB-REGION, TURBULENT DOMINANT $\Rightarrow \tau_w/\rho = (ky)^2 \cdot (d\bar{u}/dy)^2$

$\Rightarrow u_*^2 = k^2 \frac{y_+^2 \nu^2}{u_*^2} \cdot \frac{u_*^4}{\nu^2}\left(\frac{du_+}{dy_+}\right)^2$

$\Rightarrow k y_+ \frac{du_+}{dy_+} = 1 \Rightarrow u_+ = \frac{1}{k}\ln(y_+) + C$ } TURB. DOMINATES, $y_+ > 30$.

$k \approx 0.4$, $C \approx 5.0$

#. AS n INCREASES, THERE IS MORE MOMENTUM TRANSFER B/W ADJACENT FLUID LAYERS $\Rightarrow$ LESSER VALUE OF VELOCITY GRADIENT.

- FIRST LAW OF THERMODYNAMICS: $\frac{D E_{SYS.}}{Dt} = \dot{Q} - \dot{W}$

$\Rightarrow$ FROM RTT & STEADY STATE PIPE FLOW, $\dot{Q} - \dot{W} = \int_{CS} e\rho\, \vec{V}\cdot d\vec{A}$

- FROM BASIC CONSIDERATIONS, $\dot{W} = \dot{W}_{SHAFT} + \dot{W}_{PV}$, $\dot{W}_{PV} = \dot{m} pv$: v = SPECIFIC VOLUME.

$\Rightarrow \dot{Q} - \dot{W}_{SHAFT} = \int_{CS} (e + pv)\rho \vec{V}\cdot d\vec{A}$, $e = u + \frac{V^2}{2} + gz$: e = SPECIFIC INTERNAL ENERGY.

- LAMINAR FLOW, $\Delta p = \frac{128\mu L \bar{Q}}{\pi D^4} = 32\left(\frac{L}{D}\right)\left(\frac{\mu}{D}\right)\left(\frac{4\bar{Q}}{\pi D^2}\right) = 32\left(\frac{L}{D}\right)\left(\frac{\mu \bar{V}}{D}\right) \Rightarrow h_l = \frac{\Delta p}{\rho g} = \frac{\bar{V}^2}{2}\left(\frac{L}{D}\right)\left(\frac{\mu}{\rho D \bar{V}}\right)$
- TURBULENT FLOW, $\Delta p = f(D, L, e, \bar{V}, \rho, \mu) \Rightarrow \frac{\Delta p}{\rho V^2} = f\left(\frac{\mu}{\rho \bar{V} D}, \frac{L}{D}, \frac{e}{D}\right) \Rightarrow \frac{h_l}{\bar{V}^2} = \phi(Re, L/D, e)$

FROM EXPERIMENTS, VARIATION W.R.T. L/D IS LINEAR

$\Rightarrow \frac{h_l}{\frac{1}{2}\bar{V}^2} = \frac{L}{D} f_2(Re, e/D) \Rightarrow h_l = f\frac{L}{D}\cdot\frac{\bar{V}^2}{2}$

- EMPIRICAL CORRELATIONS: (COLEBROOK) $\frac{1}{\sqrt{f}} = -2.0 \log\left\{\frac{e/D}{3.7} + \frac{2.51}{Re\sqrt{f}}\right\}$

CHAALAND) $\frac{1}{\sqrt{f}} = -1.8 \log\left\{\left(\frac{e/D}{3.7}\right)^{1.11} + \frac{6.9}{Re}\right\}$

#. FOR DIFFUSERS, WE ARE USUALLY GIVEN THE PRESSURE RECOVERY COEFFICIENT. $C_p = \frac{p_2 - p_1}{\frac{1}{2}\rho \bar{V}_1^2}$

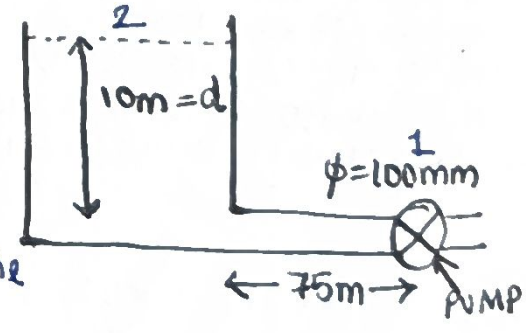
- FOR $\alpha = 1$ (TURBULENT), $\left(p_1/\rho + \frac{1}{2}\bar{V}_1^2\right) - \left(p_2/\rho + \frac{1}{2}\bar{V}_2^2\right) = h_{lm}$ (ASSUMING CONSTANT HEIGHT)

$\Rightarrow h_{lm} = \frac{\bar{V}_1^2}{2}\left\{\left(1 - \frac{\bar{V}_2^2}{\bar{V}_1^2}\right) - \frac{(p_2 - p_1)}{\frac{1}{2}\rho \bar{V}_1^2}\right\} = \frac{\bar{V}_1^2}{2}\left\{\left(1 - \frac{\bar{V}_2^2}{\bar{V}_1^2}\right) - C_p\right\}$

CONTINUITY $\Rightarrow A_1\bar{V}_1 = A_2\bar{V}_2 \Rightarrow h_{lm} = \frac{\bar{V}_1^2}{2}\left\{\left(1 - \left(\frac{A_1}{A_2}\right)^2\right) - C_p\right\} = \frac{\bar{V}_1^2}{2}\left\{\left(1 - \frac{1}{AR^2}\right) - C_p\right\}$

DEFINE $C_{pi} = 1 - \frac{1}{AR^2}$ (NO HEAD LOSS) $\Rightarrow h_{lm} = \frac{\bar{V}_1^2}{2}\{C_{pi} - C_p\}$

EXAMPLE: A 75m LONG HORIZONTAL SMOOTH PIPE IS ATTACHED TO A LARGE RESERVOIR. A PUMP IS ATTACHED TO THE END OF THE PIPE TO PUMP WATER INTO THE RESERVOIR AT THE RATE OF $0.05\ m^3/s$. WHAT PRESSURE MUST PUMP PRODUCE AT THE PIPE FOR THIS FLOW RATE. $\phi = 100mm$.



FROM CONSERVATION OF ENERGY, $\left(p_1/\rho + \alpha_1\bar{V}_1^2/2 + gh_1\right) - \left(p_2/\rho + \alpha_2\bar{V}_2^2/2 + gh_2\right) = h_l$

WHERE, p_1 = PUMP GAUGE, $p_2 = 0$, $\alpha_1 \approx 1.0$ (TURBULENT), $\bar{V}_2 \approx 0$

$\Rightarrow \frac{p_1}{\rho} + \frac{\bar{V}_1^2}{2} - gd = h_l$ (TOTAL LOSSES)

MAJOR LOSS = FLOW IN PIPE (FROM FRICTION FACTOR)

MINOR LOSS = SUDDEN EXPANSION (FROM TABLES, $K \approx 1.0$)

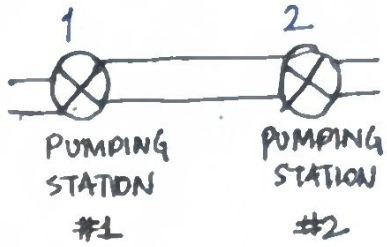
$\Rightarrow p_1/\rho + \cancel{\bar{V}_1^2/2} - gd = f\cdot(L/D)\cdot\bar{V}_1^2/2 + \cancel{\bar{V}_1^2/2} \Rightarrow p_1 = \rho\{gd + f\,L/D\cdot V_1^2/2\}$

FROM GIVEN, $Q = 0.05 \Rightarrow A = \pi\times(\frac{0.1}{2})^2 \Rightarrow \bar{V}_1 = 6.36$ m/s $\Rightarrow Re = \rho\bar{V}D/\mu = 635364$, $e = 0$ (SMOOTH)

$\Rightarrow f = 0.0126$

$\Rightarrow p_1 = 999\times\{9.81\times 10 + 0.0126\times\frac{75}{0.1}\times\frac{6.36^2}{2}\} = 289$ kPa (GAGE)

$Q = 2.944\ m^3/s$



EXAMPLE: CRUDE OIL FLOWS THROUGH A LEVEL SECTION OF THE ALASKAN PIPELINE AT A RATE OF 2.944 m^3/s. THE PIPE INSIDE DIAMETER IS 1.22m; ITS ROUGHNESS IS EQUIVALENT TO GALVANIZED IRON. MAXIMUM ALLOWABLE PRESSURE IS 8.27MPa, MINIMUM PRESSURE TO KEEP DISSOLVED GASES IN SOLUTION IS 344.5kPa. THE OIL HAS SG = 0.93; ITS VISCOSITY AT PUMPING TEMPERATURE OF 60°C IS $\mu = 0.0168\ Ns/m^2$. FOR THESE CONDITIONS, DETERMINE MAX. POSSIBLE SPACING B/W PUMPING STATIONS. IF PUMP EFFICIENCY IS 85%, DETERMINE POWER TO BE SUPPLIED AT EACH PUMPING STATION.

FROM $\rho = 0.93\times1000$, $\mu = 0.0168$, $D = 1.22$m, $\bar{V} = \frac{2.944}{\pi\times 0.61^2} \Rightarrow Re = \frac{\rho\bar{V}D}{\mu} = 1.71\times10^5$ (TURBULENT FLOW)

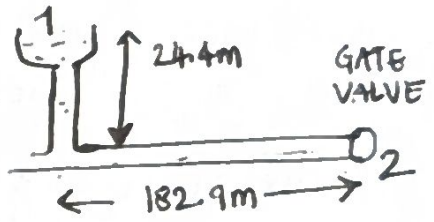
$\left(p_1/\rho + \alpha_1\bar{V}_1^2/2 + gz_1\right) - \left(p_2/\rho + \alpha_2\bar{V}_2^2/2 + gz_2\right) = h_{e,MAJOR}$ (MINOR LOSSES ARE NEGLIGIBLE)

$\Rightarrow p_1/\rho - p_2/\rho = h_{e,MAJOR} = f\cdot L/D\cdot\bar{V}^2/2 \Rightarrow L = \frac{2D}{f\bar{V}^2\rho}\{p_1 - p_2\} \approx 193$ km

PUMPING POWER HAS TO OVERCOME THIS HEAD LOSS, $\dot{W}_{PUMP} = Q\times\Delta p_{PUMP}$. HOWEVER, $\eta = 85\%$

$\Rightarrow 0.85\times\dot{W}_{INPUT} = 2.944\times(8.27\times10^6 - 3.44\times10^5) \Rightarrow \dot{W}_{INPUT} = 27.21$ MW.

EXAMPLE: A FIRE PROTECTION SYSTEM IS SUPPLIED FROM A WATER TOWER STANDPIPE 24.4 m TALL. THE LONGEST PIPE IN THE SYSTEM IS 182.9m AND IS MADE OF CAST IRON ≈ 20Y OLD. THE PIPE CONTAINS ONE GATE VALVE; OTHER MINOR LOSSES MAY BE NEGLECTED. PIPE DIAMETER IS 101.6 mm. DETERMINE MAX. RATE OF FLOW THROUGH THIS PIPE.



$\left(\cancel{p_1/\rho}^{ATM} + \alpha_1\bar{V}_1^2/2 + gz_1\right) - \left(\cancel{p_2/\rho}^{ATM} + \alpha_2\bar{V}_2^2/2 + \cancel{gz_2}^{0}\right) = h_{e,MAJOR} + h_{e,MINOR}$

$\Rightarrow g(z_1 - z_2) - \alpha_2\bar{V}_2^2/2 = f\cdot L/D\times\bar{V}_2^2/2 + K\cdot\bar{V}_2^2/2$ ($K = 0.2$ FOR GATE VALVE)

$\Rightarrow 24.4\times9.81 = \bar{V}_2^2\{\frac{f}{2}\cdot\frac{L}{D} + 0.1 + \alpha_2/2\}$ ANSATZ: ASSUME TURBULENT FLOW, & CHECK IF RESULTS ARE SELF-CONSISTENT

$\Rightarrow$ (ANSATZ) $24.4\times9.81 = \bar{V}_2^2\{\frac{f}{2}\times\frac{182.9}{0.1016} + 0.1 + 0.5\}$

FRICTION FACTOR NEEDS A REYNOLDS NO., WHICH WE DON'T KNOW. HENCE, WE ITERATE HERE. AS SOON AS WE GET SOMETHING THAT IS CONSISTENT, WE STOP ITERATING.

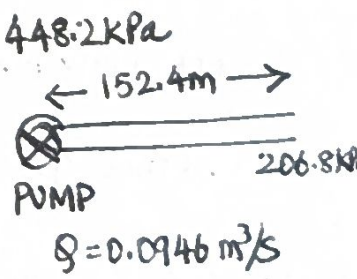
$e = 0.26$ (CAST IRON), DOUBLE IT AS OLD $\Rightarrow e/D \approx 0.005$, ASSUME (1) $f = 0.03$

$\Rightarrow 24.4\times9.81 = \bar{V}_2^2\{0.015\times182.9/0.1016 + 0.6\} \Rightarrow \bar{V}_2 \approx 2.95$ m/s $\Rightarrow Re = \frac{2.95\times0.1016}{1.124\times10^{-6}} \approx 2.66\times10^5$ (TURBULENT SATISFIED)

FOR $Re \approx 2.66\times10^5$, $e/D \approx 0.05$, $f \approx 0.003 \Rightarrow$ OUR GUESS WAS OKAY.

$\therefore Q_{MAX} = V_2\times\pi D^2/4 \approx 2.66\times10^5\times\pi\times\frac{0.1016^2}{4} \approx 0.02\ m^3/s$.

EXAMPLE: SPRAY HEADS IN AN AGRICULTURAL SPRAYING SYSTEM ARE TO BE SUPPLIED WITH WATER THROUGH 152.4m OF DRAWN ALUMINIUM TUBING FROM AN ENGINE DRIVEN PUMP. IN ITS MOST EFFICIENT OPERATING RANGE, PUMP OUTPUT IS $0.0946\ m^3/s$ AT A DISCHARGE PRESSURE NOT EXCEEDING 448.2kPa. FOR SATISFACTORY OPERATION, THE SPRINKLERS MUST OPERATE AT 206.8 kPa (GAGE), OR HIGHER. MINOR LOSSES & ELEVATION CHANGES MAY BE NEGLECTED. DETERMINE THE SMALLEST STANDARD PIPE THAT CAN BE USED.



CONSERVATION OF ENERGY, $\{p_1/\rho - p_2/\rho\} = f\,L/D\,\bar{V}^2/2$, D IS UNKNOWN.

$\Rightarrow \Delta p/\rho = f\cdot L/D\cdot\bar{V}^2/2 = f\cdot\frac{L}{D}\left(\frac{4Q}{\pi D^2}\right)^2 = \frac{8fLQ^2}{\pi^2D^5} \Rightarrow \Delta p = \frac{8\rho Q^2fL}{\pi^2D^5}$

WHERE, $Re = \frac{\rho\bar{V}D}{\mu} = \frac{4Q}{\pi\nu D}$. AT THIS POINT, WE NEED TO START ITERATING.

(i) $D^{(1)} = 0.1$m (100mm), $e = 0.0015$ mm $\Rightarrow e/D = 1.5\times10^{-5}$, $Re = \frac{4\times0.0946}{\pi\times1.21\times10^{-6}\times0.1}$ ..... KEEP GOING TILL THE PRESSURE DROP CRITERIA IS SATISFIED.